

Intermediate Macro

Chapter 3: The Goods Market

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Key Tools and Concepts

- Chapter 3 introduces **functional notation**.
- Chapter 3 introduces modeling terminology: **exogenous** and **endogenous variables**, **behavioral equations**, **identities**, and **equilibrium conditions**.
- Chapter 3 describes the **Keynesian cross** model, and associated terms, such as the (marginal) **propensity to consume**, **disposable income**, and **autonomous spending**.
- Chapter 3 introduces **fiscal policy**.

Assumptions

- The economy produces a single good and is closed (until Chapter 17).
- For short-run analysis, we assume that production adjusts to output without changes in price (**fixed price level**) (until Chapter 7).
- Within the short-run context, we assume that investment does not respond to the interest rate (This assumption will be relaxed in Chapter 5).
- In fact, we assume further that investment is **exogenous**. It does not depend on output, nor is there inventory investment (Chapter 5 introduces the dependence of investment on output).

The Composition of GDP

3.1 The Composition of GDP

- On the expenditure side, GDP can be decomposed into consumption (C), government spending (G), fixed investment (I), net exports ($X - IM$), and inventory investment.

Table 3-1 The Composition of US GDP, 2018

		Billions of Dollars	Percent of GDP
	GDP (Y)	20,500	100.0
1	Consumption (C)	13,951	68.0
2	Investment (I)	3,595	17.5
	Nonresidential	2,800	13.6
	Residential	795	3.8
3	Government spending (G)	3,522	17.2
4	Net exports	-625	-3.0
	Exports (X)	2,550	12.4
	Imports (IM)	-3,156	-15.4
5	Inventory investment	56	0.2

Source: Survey of Current Business, February 2019, Table 1-1-5

The Demand for Goods

3.2 The Demand for Goods

- Assume there is only one good, and use the decomposition of GDP to think about demand for that good.
- Assume the economy is closed, so that the net exports are zero, and ignore inventory investment.
- Demand(Z) can be written as

$$Z \equiv C + I + G.$$

- Let's discuss each of these three components in turn.

Consumption (C)

3.2 The Demand for Goods

- Consumption decisions mainly depend on income, or, more precisely, **disposable income** (Y_D).

$$C = C(Y_D) \quad (3.1)$$

(+)

$$C = c_0 + c_1 Y_D \quad (3.2)$$

where c_0 denotes **autonomous consumption** and c_1 **marginal propensity to consume**.

Consumption Function

3.2 The Demand for Goods

- We need to define disposable income Y_D .

$$Y_D \equiv Y - T$$

- Replacing Y_D in equation (3.2) gives

$$C = c_0 + c_1(Y - T) \quad (3.3)$$

Investment (I), Government Spending (G)

3.2 The Demand for Goods

- We will take investment as given **exogenous variable**.

$$I = \bar{I} \quad (3.4)$$

- In Chapter 5 we will introduce a more realistic treatment of investment.
- Together with taxes T , G describes **fiscal policy**—the choice of taxes and spending by the government.
- We will take G and T as exogenous.

Equilibrium in the Goods Market

3.3 The Determination of Equilibrium Output

- Let's put together the pieces we have introduced so far.
- We know

$$Z \equiv C + I + G$$

- Replacing C and I from equation (3.3) and (3.4), we get

$$Z = c_0 + c_1(Y - T) + \bar{I} + G \quad (3.5)$$

- Equilibrium in the goods market requires that production Y be equal to the demand for goods Z :

$$Y = Z \quad (3.6)$$

Equilibrium Condition

3.3 The Determination of Equilibrium Output

- Replacing demand Z in (3.6) by its expression from equation (3.5) gives

$$Y = c_0 + c_1(Y - T) + \bar{I} + G \quad (3.7)$$

- *In equilibrium, production Y (the left side of the equation), is equal to demand (the right side). Demand in turn depends on income Y , which is itself equal to production.*

Solving a Model

3.3 The Determination of Equilibrium Output

- Having constructed a model, we can solve it to look at what determines the level of output.
- Solving a model means not only solving it algebraically but also understanding why the results are what they are.
- Macroeconomists always use three tools:
 - ① Algebra to make sure that the logic is correct,
 - ② Graphs to build the intuition, and
 - ③ Words to explain the results.

Using Algebra

3.3 The Determination of Equilibrium Output

- Rewrite the equilibrium condition (3.7):

$$Y = c_0 + c_1 Y - c_1 T + \bar{I} + G$$

- Move $c_1 Y$ to the left side and reorganize the right side:

$$(1 - c_1)Y = c_0 + \bar{I} + G - c_1 T$$

- Divide both sides by $(1 - c_1)$:

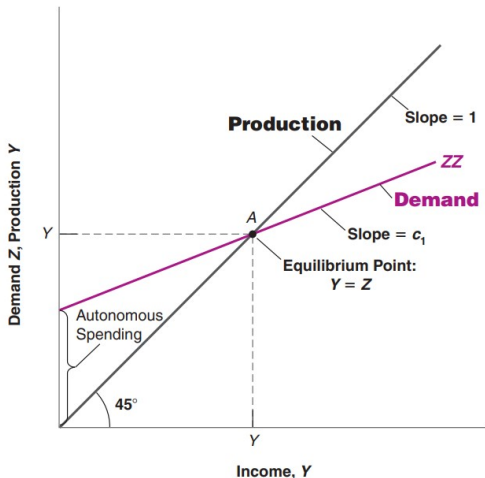
$$Y^* = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T] \quad (3.8)$$

- $\frac{1}{1 - c_1}$ is called the **multiplier**.

Using a Graph

3.3 The Determination of Equilibrium Output

- Let's characterize the equilibrium graphically.

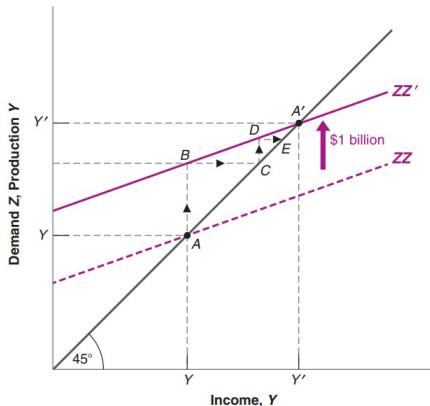


The Effects of an Increase in Autonomous Spending

3.3 The Determination of Equilibrium Output

- Rewriting equation (3.5) for convenience, we have

$$Z = (c_0 + \bar{I} + G - c_1T) + c_1Y \quad (3.9)$$



Using Words

3.3 The Determination of Equilibrium Output

- How can we summarize our findings in words?

Using Words

3.3 The Determination of Equilibrium Output

- How can we summarize our findings in words?
- Production depends on demand, which depends on income, which is itself equal to production. An increase in demand leads to an increase in production and a corresponding increase in income. This increase in income leads to a further increase in demand, which leads to a further increase in production, and so on. The end result is an increase in output that is larger than the initial shift in demand, by a factor equal to the multiplier.
- The size of the multiplier is directly related to the value of the propensity to consume. In general, it is estimated by econometrics methods.

An Alternative Way of Thinking about Equilibrium

3.4 Investment Equals Saving

- Thus far, we have been thinking of equilibrium in goods market in terms of the equality of the production and the demand for goods.
- An equivalent way of thinking about equilibrium focuses instead on investment and saving.
- Recall that production must be equal to demand:

$$Y = C + I + G$$

- Subtract taxes (T) from both sides and move consumption to the left side:

$$\underbrace{Y - T}_{\text{disposable income}} - C = I + G - T$$

$$Y_D - C = I + G - T$$

Rearranging the Equation

3.4 Investment Equals Saving

- Note that by definition, **private saving** (S) is equal to their disposable income minus their consumption. Also, by definition, **public saving** ($T - G$) is equal to taxes minus government spending.
- Then, the left side of the previous equation is simply private saving (S):

$$S = I + G - T$$

$$S = I - (T - G)$$

- Move $-(T - G)$ to the left side:

$$S + (T - G) = I$$

IS Relation

3.4 Investment Equals Saving

- Or, equivalently,

$$I = \underbrace{S + (T - G)}_{\text{total saving}} \quad (3.10)$$

- This way of looking at equilibrium explains why the equilibrium condition for the goods market is called the **IS relation**.
- Note that *consumption and saving decisions are one and the same: Once consumers have chosen one, the other is determined.*

$$\begin{aligned} S &= Y - T - C \\ &= Y - T - (c_0 + c_1(Y - T)) \end{aligned}$$

Rearranging, we get

$$S = -c_0 + (1 - c_1)(Y - T) \quad (3.11)$$

Two Equivalent Ways of Stating the Equilibrium Condition

3.4 Investment Equals Saving

- In equilibrium, investment must be equal to saving, the sum of private and public saving.
- Replacing private saving in equation (3.10) by its expression in equation (3.11),

$$I = -c_0 + (1 - c_1)(Y - T) + (T - G)$$

- Solving for output,

$$Y^* = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T]$$

which is exactly the same as equation (3.8).

Fiscal Policy Issues

3.5 Is the Government Omnipotent?

- Recall the equation (3.8):

$$Y^* = \frac{1}{1 - c_1} [c_0 + \bar{I} + G - c_1 T] \quad (3.8)$$

- The equilibrium output condition (3.8) seems to imply that the government, by choosing G and T , has absolute control over the level of output.
- Can governments really achieve the level of output they want? Obviously not.
 - 1 In reality, changing G or T is not that simply. Getting the US Congress to pass bills always takes time.
 - 2 Cutting taxes or increasing government spending comes with costs.